

## EXPERIMENT – 2

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### USER INTERFACE AND DESIGN

Design and Implementation of CLI, GUI and VUI for a Student Grade Calculation System

#### **AIM:**

To design and develop applications using GUI, CUI, and VUI interfaces for understanding different user interaction methods.

#### **COMMAND LINE INTERFACE:**

PYTHON CODE:

```
print("STUDENT PERFORMANCE EVALUATION SYSTEM")

name = input("Enter Student Name: ")

sub1 = int(input("Enter marks for Subject 1: "))
sub2 = int(input("Enter marks for Subject 2: "))
sub3 = int(input("Enter marks for Subject 3: "))
sub4 = int(input("Enter marks for Subject 4: "))
sub5 = int(input("Enter marks for Subject 5: "))

total = sub1 + sub2 + sub3 + sub4 + sub5

average = total / 5

percentage = (total / 500) * 100

print("\n----- RESULT----- ")

print("Student Name:", name)

print("Total Marks:", total)

print("Average Marks:", average)

print("Percentage:", percentage, "%")

if percentage >= 90:

    print("Grade: A+")

elif percentage >= 75:

    print("Grade: A")

elif percentage >= 60:

    print("Grade: B")

elif percentage >= 40:

    print("Grade: C")
```

else:

```
print("Grade: FAIL")
```

OUTPUT:

```
PS C:\Users\user\Desktop\GradeCal> python grade_cli.py
STUDENT PERFORMANCE EVALUATION SYSTEM
Enter Student Name: Theju
Enter marks for Subject 1: 75
Enter marks for Subject 2: 94
Enter marks for Subject 3: 62
Enter marks for Subject 4: 86
Enter marks for Subject 5: 98

----- RESULT -----
Student Name: Theju
Total Marks: 415
Average Marks: 83.0
Percentage: 83.0 %
Grade: A
```

### **GRAPHICAL USER INTERFACE:**

PYTHON CODE:

```
import tkinter as tk

from tkinter import messagebox
def calculate():

    try:

        name = name_entry.get()

        s1 = int(sub1_entry.get())

        s2 = int(sub2_entry.get())

        s3 = int(sub3_entry.get())

        s4 = int(sub4_entry.get())

        s5 = int(sub5_entry.get())

        total = s1 + s2 + s3 + s4 + s5
```

```

percentage = (total / 500) * 100
if percentage >= 90:

    grade = "A+"

elif percentage >= 75:

    grade = "A"

elif percentage >= 60:

    grade = "B"

elif percentage >= 40:

    grade = "C"

else:

    grade = "FAIL"
result = f"Name: {name}\nTotal: {total}\nPercentage: {percentage:.2f}%\nGrade:
{grade}"

messagebox.showinfo("Result", result)

except:

    messagebox.showerror("Error", "Please enter valid marks!")

root = tk.Tk()

root.title("Student Performance Evaluation System")
tk.Label(root, text="Student Name").pack()

name_entry = tk.Entry(root)

name_entry.pack()

tk.Label(root, text="Subject 1").pack()

sub1_entry = tk.Entry(root)

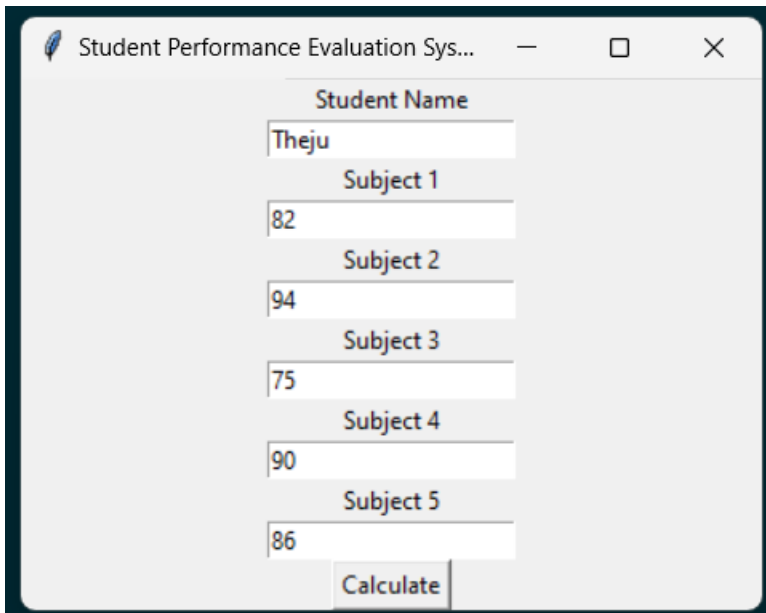
sub1_entry.pack()

tk.Label(root, text="Subject 2").pack()

```

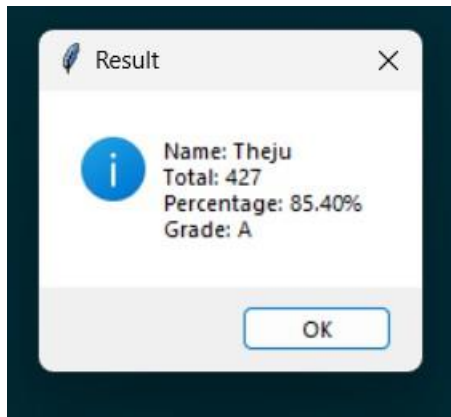
```
sub2_entry = tk.Entry(root)
sub2_entry.pack()
tk.Label(root, text="Subject 3").pack()
sub3_entry = tk.Entry(root)
sub3_entry.pack()
tk.Label(root, text="Subject 4").pack()
sub4_entry = tk.Entry(root)
sub4_entry.pack()
tk.Label(root, text="Subject 5").pack()
sub5_entry = tk.Entry(root)
sub5_entry.pack()
tk.Button(root, text="Calculate", command=calculate).pack()
root.mainloop()
```

OUTPUT:



The screenshot shows a Tkinter window titled "Student Performance Evaluation Sys...". Inside the window, there is a form with the following elements:

- A label "Student Name" above an entry field containing the text "Theju".
- A label "Subject 1" above an entry field containing the value "82".
- A label "Subject 2" above an entry field containing the value "94".
- A label "Subject 3" above an entry field containing the value "75".
- A label "Subject 4" above an entry field containing the value "90".
- A label "Subject 5" above an entry field containing the value "86".
- A "Calculate" button at the bottom right of the form.



### **VOICE USER INTERFACE:**

PYTHON CODE:

```
import speech_recognition as sr
import pyttsx3
engine = pyttsx3.init()
recognizer = sr.Recognizer()
def speak(text):
    print("System:", text)
    engine.say(text)
    engine.runAndWait()
def listen():
    with sr.Microphone() as source:
        print("Listening...")
        recognizer.adjust_for_ambient_noise(source)
        audio = recognizer.listen(source)
    try:
        text = recognizer.recognize_google(audio)
        print("You said:", text)
```

```

        return text
    except sr.UnknownValueError:
        speak("I did not understand. Please say again.")
        return listen()
    except sr.RequestError:
        speak("Internet connection problem.")
        return "0"

def get_marks(subject_number):
    while True:
        speak(f"Say marks for subject {subject_number}")
        value = listen()
        try:
            marks = int(value)
            if 0 <= marks <= 100:
                return marks
            else:
                speak("Marks should be between 0 and 100.")
        except:
            speak("Please say the number clearly.")

speak("Welcome to Student Performance Evaluation System")
speak("Say student name")
name = listen()
marks_list = []

```

```
for i in range(1, 6):  
    mark = get_marks(i)  
    marks_list.append(mark)  
total = sum(marks_list)  
percentage = (total / 500) * 100  
if percentage >= 90:  
    grade = "A plus"  
elif percentage >= 75:  
    grade = "A"  
elif percentage >= 60:  
    grade = "B"  
elif percentage >= 40:  
    grade = "C"  
else:  
    grade = "Fail"  
result = f"{name} scored {percentage:.2f} percent. Grade is {grade}"  
print("\n----- RESULT----- ")  
print("Student Name:", name)  
print("Total Marks:", total)  
print("Percentage:", percentage)  
print("Grade:", grade)  
speak(result)
```

OUTPUT:

```
System: Welcome to Student Performance Evaluation System
System: Say student name
Listening...
System: I did not understand. Please say again.
Listening...
You said: Sia
System: Say marks for subject 1
Listening...
You said: 92
System: Say marks for subject 2
Listening...
You said: 86
System: Say marks for subject 3
Listening...
You said: 75
System: Say marks for subject 4
Listening...
You said: 98
System: Say marks for subject 5
Listening...
You said: 78

----- RESULT -----
Student Name: Sia
Total Marks: 429
Percentage: 85.8
Grade: A
System: Sia scored 85.80 percent. Grade is A
PS C:\Users\user\Desktop\GradeCal>
```

### **USER SATISFACTION COMPARISON:**

PYTHON CODE:

```
print("USER SATISFACTION SURVEY")

print("1 - Very Bad")

print("2 - Bad")

print("3 - Average")

print("4 - Good")

print("5 - Excellent")

cli = int(input("Rate CLI (1-5): "))

gui = int(input("Rate GUI (1-5): "))

vui = int(input("Rate VUI (1-5): "))
```

```

avg = (cli + gui + vui) / 3
print("\nSurvey Results")

print("CLI Rating :", cli)

print("GUI Rating :", gui)

print("VUI Rating :", vui)

print("Average Satisfaction Score =", round(avg, 2))
if avg >= 4:

    print("Overall Feedback: Excellent")

elif avg >= 3:

    print("Overall Feedback: Good")

elif avg >= 2:

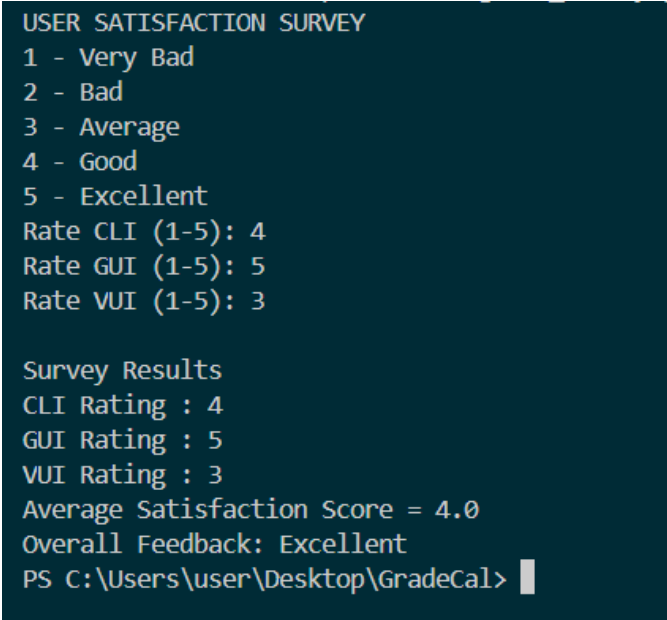
    print("Overall Feedback: Average")

else:

    print("Overall Feedback: Poor")

```

OUTPUT:



```

USER SATISFACTION SURVEY
1 - Very Bad
2 - Bad
3 - Average
4 - Good
5 - Excellent
Rate CLI (1-5): 4
Rate GUI (1-5): 5
Rate VUI (1-5): 3

Survey Results
CLI Rating : 4
GUI Rating : 5
VUI Rating : 3
Average Satisfaction Score = 4.0
Overall Feedback: Excellent
PS C:\Users\user\Desktop\GradeCal>

```

## RESULT:

The GUI and CUI calculated grades successfully, while the VUI successfully performed arithmetic operations using voice commands.